Bukang-liwayway v3.4 — Printing & Assembly Guide

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Before you start

Bukang-liwayway ("Sunrise") is Sol Prosthetics' Model 1: a wrist-driven hand for children about 9 and under. Bending the wrist down pulls four braided lines and the fingers close; bending it back lets elastic cords open them. The thumb clicks into one of three positions (open, middle or pinch, 30° apart) and is locked with a knob you turn by hand, no tools. The palm side is a soft TPU cover screwed on all round, and the child's hand slides in through the wrist end like a glove.

!! SAFETY: ALWAYS WEAR EYE PROTECTION

Wear safety glasses whenever you build or finish the parts. Snipping supports and necks, drilling, trimming and cutting line can flick sharp bits of plastic into your eyes. The soldering iron for the inserts runs at about 240 °C: keep it in its stand and never leave it on alone. Print, set inserts and glue (E6000) in a room with fresh air. Keep cutters, the iron and loose screws away from small children: small parts are a choking risk.

- **This is a test print.** v3.4 is the fourth test version: the v3.3 design with the thumb mount braced solid to the palm, bigger thumb click teeth, and a support blocker that keeps supports out of the thumb hinge. It has not been fitted to a child yet. Treat it as a trial and check everything in the Tuning section.
- **Print at 100% only.** Never scale the files in the slicer: the screw holes, insert holes and joint gaps would change size. Ready-made size sets, each with its own hardware card, come after the test fit.
- **It fits** a child's hand about 55 mm across the knuckles and 129 mm from wrist crease to middle fingertip, with a palm up to about 49 mm long from the wrist crease. Wrist about 97 mm round, forearm about 110 mm.
- **Materials:** PETG for every rigid part (never PLA, it softens in the heat) and TPU 95A for the palm cover and the optional art holder.
- If you can, fit the first hands with a physical therapist or prosthetist. Every residual limb is different.
- **Small parts and fiddly steps.** Many parts are small and need careful finishing: snipping supports and thin necks, hand-drilling holes, setting hot brass inserts with a soldering iron, threading 0.32 mm line through 2 mm holes, and tightening small screws. If you're building the hand yourself, for example a teenager printing their own, and your grip or fine finger control is limited, ask someone to help with these steps. Children should always have an adult for the soldering iron and the cutters.

Parts to print

Print one of each part below from the `stl` folder (GitHub or the website). Every file is already turned to its print direction, so load it as it is and don't rotate it. Print the two `_print` versions: they have snip-off supports built in.

File	Part	Material	Size (mm)	Supports
`palm_back_print.stl`	Back of the hand, with a snip-off rim support	PETG	74 × 52 × 77	Normal + its blocker; brim
`palm_flap_print.stl`	Palm cover, with snip-off fins and sole bar	**TPU 95A**	71 × 25 × 49	Build plate only; brim

File	Part	Material	Size (mm)	Supports
`cuff.stl`	Forearm cuff	PETG	63 × 39 × 109	A few
`thumb.stl`	Thumb	PETG	25 × 26 × 30	Build plate only; brim
`thumb-knob.stl`	Thumb knob (clips onto the thumb mount)	PETG	15 × 15 × 7	None; 100% infill
`finger-index-1/2/3.stl`	Index finger, 3 segments	PETG	up to 18 × 30 × 30	Build plate only; brim
`finger-middle-1/2/3.stl`	Middle finger, 3 segments	PETG	up to 17 × 34 × 35	Build plate only; brim
`finger-ring-1/2/3.stl`	Ring finger, 3 segments	PETG	up to 15 × 31 × 32	Build plate only; brim
`finger-little-1/2.stl`	Little finger, 2 segments	PETG	up to 16 × 25 × 24	Build plate only; brim
`art-holder.stl`	Art holder, optional (pencil, pen, crayon, marker, brush)	**TPU 95A**	18 × 35 × 18	None

`palm_back.stl` and `palm_flap.stl` are the same parts without the print aids, for reference. `HARDWARE.md` in the same folder lists every screw and insert for this size.

Segment 1 is the one that joins the palm; segment 3 (2 on the little finger) is the fingertip with the nail. Plastic needed: roughly 100–135 g of PETG plus supports, and under 10 g of TPU.

Print settings

A 0.4 mm nozzle and 0.2 mm layers are assumed for every part.

Setting	PETG parts	TPU palm cover and art holder
Walls	5	4
Top / bottom layers	Your usual	4 each
Infill	40–50% gyroid (fingers, thumb, knob, back of hand); 30% (cuff)	30% gyroid (the 2 mm cover is almost all walls and top/bottom layers)
Speed	Normal for your PETG	Slow, 20–30 mm/s (the art holder's thin fins need it)
Retraction	Normal	Little or none
Surface	Normal	**Fuzzy skin** on the cover's outer grip face only, so the palm isn't slippery. Keep it off the 12 screw countersinks round the edge. Optional: TPU 95A is grippy anyway.
Supports	Back of the hand: normal supports with its blocker loaded (below). Fingers and thumb: on the build plate only. Knob: none. Cuff: a few	Cover: build plate only (few or none). Art holder: none
Brim	5 mm: back of the hand, fingers and thumb	5 mm (cover)

Per-part tips:

- **Screw holes never need support.** Every screw hole prints lying sideways and is small (2.6–3.4 mm, with 6.2 mm head recesses), so each layer only bridges a few millimetres over the top of the hole.
- **Fingers and thumb:** they stand at 45°, knuckle end down and nail side up, so no support scars land on the faces you see. Supports on the build plate only: that removes every support the slicer wants inside the screw holes and keeps the ones under the low knuckle end. Use a brim, because each part touches the bed only a little. The thumb knob prints as it comes in the file, with no supports. If the thumb gives trouble on a full plate, print it on its own: it has come out better that way on our test prints.
- **Back of the hand:** it stands on its wrist end, so the line tunnels run almost straight up. It touches the bed only under the two wrist ears, and the rim across the back floats about 8 mm up, right where the line tunnels come out. `palm_back_print.stl` holds that rim on 14 small posts with thin necks, all on solid plastic between the tunnel exits, standing on a thin sole that also keeps the tall part from tipping over. Keep normal supports on (the inside of the knuckles needs them) and load `palm_back_print_support_blocker.stl` with it, so no support goes into a hole, a line tunnel or the thumb hinge (its slot, the click teeth and the bolt hole). The thumb mount is joined solid to the palm, so it grows up out of the palm wall and needs no support under it. Supports never touch the logo.
  - **Creality Print / OrcaSlicer / PrusaSlicer:** drag both files in together and answer **Yes** to loading them as one object with multiple parts. Right-click the blocker part → **Change type** → **Support blocker**.
  - **Cura:** turn off "Automatically drop models to the build plate" and "Ensure models are kept apart" before loading, so both keep their positions. Select the blocker → **Per Model Settings** → **Don't support overlaps**.
- **Palm cover:** print `palm_flap_print.stl`. Dry the TPU first if it has been sitting out; wet TPU strings and blobs.
- **All parts:** every file is already a clean, closed solid. Don't run the slicer's mesh-repair, "fill holes" or "remove small features" options on it.

Art holder (optional): TPU 95A, standing on its back end as it comes in the file, no supports. Print it slowly so the six thin fins inside the tube come out clean.

Hardware and materials

All screws are stainless M3 from a hardware store, and no nuts are needed. The finger joint screws thread into 2.6 mm pilot holes in the plastic. The knuckle axle, the thumb knob's bolt, the 12 cover screws and the 2 wrist screws thread into brass heat-set inserts, 16 in all. This is the list for the 100% size; `HARDWARE.md` in the `stl` folder has the list for the size you print.

Item	Qty	Where it goes
M3 × 45 **button-head** screw (ISO 7380)	1	Knuckle axle: in from the index side, through the index, middle and ring knuckles, into the insert in the index's outer arm
M3 × 12 button-head screw	8	Middle and tip joints of the index, middle and ring fingers; the little finger's knuckle and tip joint
M3 × 20 **hex-head** bolt (DIN 933, fully threaded)	1	Thumb knob: its head is pressed into the knob; threads into the insert in the thumb
M3 × 6 button-head screw	2	Wrist hinge, one through each cuff arm into the insert in the hand's ear pad
M3 × 5 **countersunk (flat-head)** screw	12	The palm cover, all round its edge
M3 × 4 countersunk (flat-head) screw	4	Line clamps on top of the cuff

Item	Qty	Where it goes
M3 brass **heat-set insert**, 4.2 mm across × 4 mm long (double-knurled)	16	Index knuckle outer arm 1, thumb 1, cover step 12, wrist ear pads 2 (one cover hole near the thumb-side wrist corner also takes a 5 mm insert)
Blue (medium, removable) threadlocker	a drop each	Knuckle axle and the 2 wrist screws
PE braided fishing line #4, 50 lb (about 0.32 mm)	about 2 m	4 flexor lines (close the fingers)
1 mm clear round elastic cord	about 1.5 m	4 return cords (open the fingers)
Bicycle inner tube, about 1 mm (butyl)	1 small piece	5 grip pads
E6000 or contact cement	a little	Gluing the grip pads
Sew-on hook-and-loop strap, 25 mm	2	Cuff straps
Thin cotton sock or stockinette (tube gauze)	1, plus spares	The liner, worn on the hand and forearm; wash it daily

Tools you need

- 2 mm hex key (fits every button-head and flat-head screw), with a long one or a hex bit for the 45 mm knuckle axle
- Soldering iron with a heat-set insert tip (or a plain tip), set to about 240 °C for PETG
- 3.5 mm drill bit, turned by hand (a pin vise or just your fingers), for cleaning the 3.4 mm screw holes
- Flush cutters and a craft knife for removing supports, the snip-off necks and brims
- Fine sandpaper or a needle file
- Thin wire or a long darning needle for threading lines through the tunnels
- Lighter or a drop of super glue for sealing knots in the braided line
- Scissors, rubbing alcohol and a cloth
- Needle and strong thread for sewing on the straps
- Small flat screwdriver, only for easing the thumb knob's clip fingers over their rim if you take the thumb off

After printing

- ☐ **Back of the hand:** snip each of the 14 necks flush at the rim with flush cutters, lift the support and sole off, and sand the marks smooth. Check that every line-tunnel exit on the rim is clear.
- ☐ **Palm cover:** snip the small tabs off the fins and the sole bar, and shave any nub flush.
- ☐ Remove the other supports and brims. Pay attention inside the knuckle cups, the thumb mount slot and the ring of grooves in it.
- ☐ Sand smooth any edge that will touch skin: the open wrist end of the back, the cuff ends and strap slots.
- ☐ **Clean the screw holes.** The top of a hole printed without support can sag 0.1–0.2 mm. By hand, run the 3.5 mm drill bit through the 3.4 mm holes the screws slide through (finger joints, knuckle axle, and the thumb mount's screw path). **Don't drill the 2.6 mm pilot holes** the screws thread into: that would loosen the grip. Then run an M3 screw through every pilot hole once to cut the thread, and back it out.
- ☐ Push a piece of line through every channel and tunnel to check it is clear: the upper and lower channel in every finger segment, 8 tunnels in the back of the hand, 4 in the cuff.

- ☐ Check that the 12 insert holes in the cover step and the 2 in the wrist ear pads are clear.
- ☐ Dry-fit the cover in its step: it should lie flush all round with no gaps.

Assembly

Work in this order; each step leaves the next screw reachable. Tighten joints only until they move freely without wobble.

1. **Set the inserts.** Soldering iron at about 240 °C for PETG (350 °C is too hot and melts the hole oversize). Let the iron's weight push each insert in, then press it flush with something flat and cool. There are 16, and each goes in from outside and finishes flush: one in the index knuckle segment's outer arm (the thumb side, under the axle head), one in the thumb's tongue (from the face with the ring of teeth), 12 in the cover step all round the opening, and one in each wrist ear pad of the back of the hand. The palm has **no** insert at the thumb, just a plain hole for the bolt.
2. **Build each finger.** Join its segments in order (1 to 2, 2 to 3). Each joint is a rounded barrel in a cup held by one button-head screw. Put the screw in from the side with the round recess, so the head sits flush; it threads into the pilot hole on the far side. A flat stop on the back of each joint keeps the finger from bending backwards.
   - Index, middle and ring: M3 × 12 (middle joint), M3 × 12 (tip joint)
   - Little: M3 × 12 (tip joint)
3. **Knuckle axle.** Line up the index, middle and ring knuckle cups on their tongues on the back of the hand. Push the M3 × 45 in from the index side, through all three knuckles, and screw it into the index insert until its head sits in the recess. Put a drop of blue threadlocker on the thread first. It is a plain pin through the middle and ring knuckles, so they swing freely on it. The insert sits in the index's outer arm, under the head, so tightening can't squeeze the index onto the palm: it must still swing freely when the axle is tight.
4. **Little finger.** Slide its cup over its knuckle tongue and screw in its own M3 × 12 from the recessed side.
5. **Thumb.** Push the head of the M3 × 20 hex bolt up into the knob's hex pocket until it bottoms (warm the head with the iron if it is tight). Fit the thumb's tongue into the slot on the thumb mount, its ring of teeth toward the knob side, no washers. Push the bolt through and turn the knob to start the thread in the thumb's insert. Press the knob straight down (no twisting) until its eight fingers click over the rim on the post, then swing the thumb to the click you want and tighten the knob firmly by hand.
6. **Return cords (open the fingers).** Knot the elastic in the knot pocket under the fingertip pad. Thread it through the upper channel of each segment and through its tunnel in the back of the hand. Knot it at the wrist end, just snug enough to hold the finger open. A surgeon's knot with a drop of glue holds best in the smooth elastic.
7. **Flexor lines (close the fingers).** Knot each braided line in the same fingertip pocket and seal the knot. Thread it through the lower channel of each segment, into the knuckle block and through its tunnel to the wrist end. A thin wire or long needle helps; the holes are 2–2.5 mm with 3 mm funnel mouths.
8. **Wrist.** Join the back of the hand to the cuff with the two M3 × 6 screws through the cuff's hinge arms into the inserts in the hand's ear pads. Each screw is also the hinge pin: put a drop of blue threadlocker on it and tighten only until the cuff swings freely.
9. **Tensioner.** Feed each flexor line into its hole in the front end of the cuff. It runs inside the wall, under its clamp, and comes out on the inside of the cuff behind the swelling. With the wrist straight and the fingers open, pull it snug and screw in its M3 × 4 flat-head clamp screw from the top until it sits flush; the screw pins the line. Trim the tail inside the cuff.
10. **Grip pads.** Cut inner tube to the recess sizes below. Sand the rubber and the recess dull, wipe both with alcohol, put a thin film of E6000 or contact cement on both and wait until tacky (2–5 minutes). Press the pad in, clamp it for 24 hours, and let it cure 72 hours before a child wears the hand.
11. **Palm cover.** Lay the cover in its step and start all 12 M3 × 5 flat-head screws loosely from outside into the inserts, then tighten each until its head sits flush in the TPU. Don't overtighten: it squashes the TPU. The

cover stays on; the hand goes in through the wrist end. To change a cover, unscrew the 12 screws; nothing is glued.

12. **Straps and liner.** Sew the two straps onto the cuff through its strap slots and sand any edge that touches skin. The liner is a thin cotton sock or stockinette (tube gauze) pulled over the child's hand and forearm before the hand goes on: it soaks up sweat and is easy to wash or swap every day.

Grip pad (cut as an oval)	Cut size (mm)
Index fingertip	7 × 6.5
Middle fingertip	7 × 6.5
Ring fingertip	7 × 6.5
Little fingertip	7 × 9
Thumb tip	9.5 × 8

Art holder (optional). Slide it over the index fingertip onto the finger's middle segment, tube on the thumb side. Push a pencil, pen, crayon, marker or paintbrush into the tube from the back, point forward, with a little twist: the fins inside hold anything from about 4.5 mm (a thin brush) to 11.5 mm (a fat marker), round, hex or triangular. Set the thumb to its pinch click while it is on; in the middle click the thumb meets the tube when the fingers close.

Tuning and test checks

In CAD, the thumb (in its pinch click) meets the index fingertip at about 49–55° of finger curl, and the hand swings from about 52° down to 30° up with nothing rubbing, cover included. Check the real print against that.

- ☐ **Knuckle axle:** the index, middle and ring fingers should each swing freely on the long axle without dragging their neighbours. With the insert in the index's outer arm, tightening the axle shouldn't bind anything. If a finger still drags, look for support left in its cup.
- ☐ **Line tension:** bend the wrist down and watch all four fingers. If one closes late or not at all, loosen its clamp screw, pull its line a little tighter and re-tighten. Adjust until all four close together.
- ☐ **Opening:** with the wrist straight, every finger should spring fully open. If one stays bent, tighten its return cord knot at the wrist end.
- ☐ **Joint friction:** each joint should swing freely with no side wobble. Loosen or tighten its screw by a quarter turn at a time.
- ☐ **Pinch:** close the hand and check that the thumb meets the index fingertip. To move the thumb, turn the knob back until it stops (about 2 turns), swing the thumb one click (30°) and tighten the knob again. Tightened firmly by hand, the knob must hold the thumb in a hard pinch; if it slips, note it for the next version. The teeth are 1 mm tall in v3.4: check they printed sharp and clean on both the thumb and the palm.
- ☐ **Wrist range:** the hand should swing from about 52° down to 30° up without rubbing the cuff or the cover.
- ☐ **Cover:** the edge should lie flush all round, with all 12 screw heads flush and the TPU not puckered between them.
- ☐ **Fit:** with the liner (sock) on, the child's hand should slide in through the wrist end with the cover screwed on without forcing, and nothing should press on the skin.

Putting it on and daily care

1. Pull the liner, a thin cotton sock or stockinette, over the child's hand and forearm.

2. Slide the hand in through the wrist end, like a glove. The cover stays screwed on; the wrist end flares open a little to help.
3. Do up the two cuff straps, snug but not tight.
4. Set the thumb to the click the child needs and tighten the knob by hand.
5. **Check the child's skin every day,** especially in the heat. A closed shell hides red marks and pressure spots. If a mark doesn't fade within about 20 minutes, stop using the hand until it has been checked.
6. **Wash** the cover and the shell, and the liner sock every day, with soap and water. Let them drain and dry; a drain hole at the front of the hand lets water out.
7. **Keep it out of hot cars and direct sun** for long periods. PETG holds up far better than PLA, but heat still softens plastic.
8. **Replace worn parts** by reprinting them: each finger segment, the thumb, the knob and the cover are separate files, and the cover comes off with its 12 screws. Replace a flexor line if it frays.